Complex Number: Formula Sheet

1. Basics

Iota: $i^2 = -1,\ i^{4k+r} = i^r$

Four powers: $i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$

Division:

$$\frac{a+ib}{c+id} = \frac{(ac+bd)+i(bc-ad)}{c^2+d^2}$$

Square root:

$$\sqrt{a+ib} = \pm\left(\sqrt{\tfrac{|z|+a}{2}} + i\,\tfrac{b}{|b|}\sqrt{\tfrac{|z|-a}{2}}\right)$$

2. Conjugate and modulus

Conjugate:

$$\overline{z_1 z_2} = \bar{z}_1\bar{z}_2,\ \overline{z_1+z_2} = \bar{z}_1+\bar{z}_2$$

Real / imaginary: $z = \bar{z}$ / $z + \bar{z} = 0$

Modulus:

$$|z|^2 = z\bar{z},\ |z_1 z_2| = |z_1||z_2|$$

Expansion:

$$|z_1 \pm z_2|^2 = |z_1|^2 + |z_2|^2 \pm 2\,\mathrm{Re}(z_1\bar{z}_2)$$

Parallelogram law:

$$|z_1+z_2|^2 + |z_1-z_2|^2 = 2(|z_1|^2+|z_2|^2)$$

Triangle inequality:

$$\big||z_1|-|z_2|\big| \le |z_1+z_2| \le |z_1|+|z_2|$$

3. Argument and forms

Polar / Euler:

$$z = r(\cos\theta + i\sin\theta) = re^{i\theta}$$

Product, quotient: $\arg z_1 z_2 = \arg z_1 + \arg z_2$, $\arg \frac{z_1}{z_2} = \arg z_1 - \arg z_2$

Conjugate: $\arg \bar{z} = -\arg z$

Half angle:

$$1 + e^{i\theta} = 2\cos\tfrac{\theta}{2}e^{i\theta/2}$$

Half angle:

$$1 - e^{i\theta} = -2i\sin\tfrac{\theta}{2}e^{i\theta/2}$$

Tangent form:

$$1 + i\tan\theta = \sec\theta\, e^{i\theta}$$

DMT:

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$$

nth root:

$$z^{1/n} = r^{1/n}e^{i(\theta+2k\pi)/n}$$

4. Geometry

Distance: $|z_1 - z_2|$

Section:

$$\frac{mz_2 \pm nz_1}{m \pm n}$$

Centroid, incentre:

$$\frac{z_1+z_2+z_3}{3},\ \frac{az_1+bz_2+cz_3}{a+b+c}$$

Parallelogram: $z_1 + z_3 = z_2 + z_4$

Rotation:

$$\frac{z_3-z_1}{z_2-z_1} = \frac{|z_3-z_1|}{|z_2-z_1|}e^{i\theta}$$

Parallel / perpendicular: $\frac{z_4-z_3}{z_2-z_1}$ real / purely imaginary

Equilateral:

$$z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$$

Regular polygon:

$$z_k = z_0 + (z_1 - z_0)e^{i2\pi(k-1)/n}$$

5. Loci

Circle: $|z - z_0| = r$

Bisector / Apollonius: $\left|\frac{z-z_1}{z-z_2}\right| = 1$ / $= k \neq 1$

Ellipse: $|z-z_1| + |z-z_2| = k > |z_1-z_2|$

Hyperbola:

$$\big||z-z_1| - |z-z_2|\big| = k < |z_1-z_2|$$

Arc: $\arg\frac{z-z_1}{z-z_2} = \theta$; circle on diameter: $\theta = \pm\frac{\pi}{2}$

Line: $\bar{a}z + a\bar{z} + b = 0$; through z_1, z_2: $z = (1-t)z_1 + tz_2$

Circle: $z\bar{z} + \bar{a}z + a\bar{z} + b = 0$, centre $-a$, $r = \sqrt{|a|^2 - b}$

Image in a line: $\bar{a}z_2 + a\bar{z}_1 + b = 0$

Concyclic:

$$\frac{(z_4-z_1)(z_2-z_3)}{(z_2-z_1)(z_4-z_3)} \in \mathbb{R}$$

6. Roots of unity

Cube roots: $1,\ \omega = \frac{-1+i\sqrt{3}}{2},\ \omega^2$; $1 + \omega + \omega^2 = 0$

Powers of ω: $1 + \omega^r + \omega^{2r} = 3$ or 0

Factors:

$$x^2 \pm x + 1 = (x \mp \omega)(x \mp \omega^2)$$

Cubic identity:

$$a^3+b^3+c^3-3abc = (a+b+c)(a+b\omega+c\omega^2)(a+b\omega^2+c\omega)$$

nth roots: $e^{i2k\pi/n}$; sum 0; product $(-1)^{n-1}$

Power sums: $\sum \alpha_k^r = n$ if $n \mid r$, else 0

Factorisation:

$$\prod_{k=1}^{n-1}(x-\alpha_k) = 1 + x + \cdots + x^{n-1}$$

Real factor: $x^2 - 2x\cos\frac{2k\pi}{n} + 1$

7. Series

Cosine sum:

$$\sum_{k=0}^{n-1}\cos(\alpha+k\beta) = \frac{\sin(n\beta/2)}{\sin(\beta/2)}\cos\left(\alpha+\frac{(n-1)\beta}{2}\right)$$

Alternate:

$$C_0 - C_2 + C_4 - \cdots = 2^{n/2}\cos\frac{n\pi}{4}$$

Alternate:

$$C_1 - C_3 + C_5 - \cdots = 2^{n/2}\sin\frac{n\pi}{4}$$

Step 3:

$$C_0 + C_3 + C_6 + \cdots = \frac{1}{3}\left(2^n + 2\cos\frac{n\pi}{3}\right)$$